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|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONTACT<br>INFORMATION                     | Max-Planck-Institut für Mathematik<br>Vivatsgasse 7,<br>53111 Bonn, Germany                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Homepage: <a href="http://challeckdube.github.io">http://challeckdube.github.io</a><br>E-mail: <a href="mailto:halleckdube@mpim-bonn.mpg.de">halleckdube@mpim-bonn.mpg.de</a> |
| RESEARCH                                   | <ul style="list-style-type: none"> <li>Algebraic geometry, Representation Theory, Langlands.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                               |
| PUBLICATIONS                               | <ol style="list-style-type: none"> <li>The Weighted Fundamental Lemma for nonsplit groups; in preparation.</li> <li>(with Ascher, Gershenson, Hou) Enumerating Hassett's wall and chamber decomposition of the moduli space of weighted stable curves, <a href="#">arxiv</a>, <i>Experimental Mathematics</i>, <b>29</b>, 36-53</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                               |
| ACADEMIC<br>POSITIONS                      | <p><b>Duke University</b>, Durham, NC</p> <ul style="list-style-type: none"> <li>Griffiths Assistant Professor</li> </ul> <p><b>Max Planck Institute for Mathematics</b>, Bonn, Germany</p> <ul style="list-style-type: none"> <li>Visiting Postdoctoral Fellow</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>2027-</p> <p>2026-2027</p>                                                                                                                                                 |
| EDUCATION                                  | <p><b>University of California at Berkeley</b>, Berkeley, CA</p> <ul style="list-style-type: none"> <li>Ph.D. in Mathematics</li> <li>Thesis: "The Cohomological Weighted Fundamental Lemma for nonsplit groups"</li> </ul> <p><b>Institut de Mathématiques de Jussieu–Paris Rive Gauche</b>, Paris, FR</p> <ul style="list-style-type: none"> <li>Visiteur</li> </ul> <p><b>Cambridge University</b>, Cambridge, UK</p> <ul style="list-style-type: none"> <li>Master of Advanced Study in Mathematics</li> </ul> <p><b>Yale University</b>, New Haven, CT</p> <ul style="list-style-type: none"> <li>B.Sc. in Mathematics, GPA: 3.97/4.00, <i>summa cum laude</i>, Phi Beta Kappa</li> <li>Undergraduate Thesis: "A Database of Cubic Fourfolds over <math>\mathbb{F}_2</math>," advised by Asher Auel</li> </ul> <p><b>Oxford University</b>, Oxford, UK</p> <ul style="list-style-type: none"> <li>Registered Visiting Student</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>2026</p> <p>Fall 2023</p> <p>2020</p> <p>2019</p> <p>2017-2018</p>                                                                                                         |
| TEACHING<br>AND<br>MENTORING<br>EXPERIENCE | <ul style="list-style-type: none"> <li>UC Berkeley, Summer REU Mentor <ul style="list-style-type: none"> <li>Project: "Explicit computations of Hecke algebras and stable orbital integrals for classical groups."</li> </ul> </li> <li>UC Berkeley, Graduate Student Instructor <ul style="list-style-type: none"> <li>Spring 2026: GSI for Math 10B (Statistics and combinatorics for life sciences)</li> <li>Fall 2025: GSI for Math 1 (Precalculus)<br/><i>Partnership with National Educational Equity Lab</i><br/><i>Awarded NEON Teaching Fellow Award</i></li> <li>Spring 2025: GSI for Math 113 (Abstract Algebra)</li> <li>Fall 2024: GSI for Math 110 (Linear Algebra)</li> <li>Spring 2024: GSI for Math 110 (Linear Algebra)</li> <li>Summer 2023: Lead Instructor for Math 54 (Linear Algebra)<br/><i>Berkeley Outstanding Instructor Award 2022-2023</i></li> <li>Fall 2022: GSI for Math 55 (Discrete Mathematics)</li> <li>Spring 2021: GSI/Lecturer for Math 191 (Special Topics in Mathematics)<br/><i>A second course in linear algebra from a categorical perspective</i></li> <li>Fall 2021: GSI/Lecturer for Math 113 (Abstract Algebra)</li> <li>Spring 2021: GSI for Math 10B (Statistics and combinatorics for life sciences)</li> <li>Fall 2020: GSI for Math 10A (Calculus and differential equations for life sciences)</li> </ul> </li> <li>HCSSiM, intensive IBL residential summer mathematics program for high-school students <ul style="list-style-type: none"> <li>Junior Faculty</li> </ul> </li> <li>MathILy, intensive IBL residential summer mathematics program for high-school students <ul style="list-style-type: none"> <li>Apprentice Instructor, remote instruction</li> <li>Apprentice Instructor</li> </ul> </li> <li>Undergraduate Learning Asst., Discrete Mathematics for Computer Science, Yale</li> </ul> | <p>Summer 2025</p> <p>2024</p> <p>2020</p> <p>2019</p> <p>2018</p>                                                                                                            |

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|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| CONFERENCES,<br>PROGRAMS,<br>AND<br>WORKSHOPS | • Collaborations in Algebraic Geometry, Representation Theory, and Ethics                                                                                                               | Sept 2026   |
|                                               | • Western Algebraic Geometry Symposium                                                                                                                                                  | Fall 2025   |
|                                               | Poster: <i>Truncated Hitchin Fibrations</i>                                                                                                                                             |             |
|                                               | • Summer Research Institute in Algebraic Geometry                                                                                                                                       | Summer 2025 |
|                                               | Bootcamp group: <i>Twisted conformal blocks in algebraic geometry</i> .                                                                                                                 |             |
|                                               | Mentor: Chiara Damiolini                                                                                                                                                                |             |
|                                               | • Arizona Winter School: Representations of $p$ -adic groups                                                                                                                            | Spring 2025 |
|                                               | Project group: <i>Characters of Fintzen-Schwein representations for <math>p = 2</math></i> .                                                                                            |             |
|                                               | Mentor: Charlotte Chan                                                                                                                                                                  |             |
|                                               | • IHES Summer School in the Langlands Program                                                                                                                                           | Summer 2022 |
| TALKS                                         | • Berkeley RTG Research Workshop on Chabauty-Coleman and Rational Points                                                                                                                | Spring 2021 |
|                                               | • Arizona Winter School 2022: Automorphic Forms Beyond $GL_2$ .                                                                                                                         |             |
|                                               | • Summer Undergraduate Mathematics Research at Yale                                                                                                                                     | Summer 2017 |
|                                               | <i>Combinatorics of moduli of weighted stable curves</i> . Advised by Kenneth Ascher.                                                                                                   |             |
|                                               | • Summer Undergraduate Mathematics Research at Yale                                                                                                                                     | Summer 2016 |
|                                               | <i>Divisors on <math>\overline{M}_{0,n}</math></i> . Advised by Jose Gonzalez.                                                                                                          |             |
|                                               | • <i>Geometry of the Weighted Fundamental Lemma</i> . Invited Talk. Algebraic Geometry and Number Theory seminar, UC Davis, 4 Feb. 2026.                                                |             |
|                                               | • <i>The Weighted Fundamental Lemma and Geometric Endoscopy</i> . Invited Talk. Junior Number Theory Days, Johns Hopkins University, 31 Jan. 2026.                                      |             |
|                                               | • <i>The Weighted Fundamental Lemma for nonsplit groups</i> . Invited Talk. Workshop on Arthur's Endoscopic Classification, University of Chicago Paris Center, 9 Sept. 2025.           |             |
|                                               | • <i>Hitchin Truncation for Relative Reductive Groups</i> . Spec $\overline{\mathbb{Q}}(2\pi i)$ , 19 June 2024.                                                                        |             |
| FELLOWSHIPS<br>AND AWARDS                     | • <i>The Cartier Isomorphism and De Rham Cohomology in Characteristic <math>p</math></i> . Number Theory Learning Seminar, UC Berkeley, 12 Oct. 2022.                                   |             |
|                                               | • <i>Moduli Space of Langlands Parameters</i> . Berkeley-Stanford Joint Number Theory Working Seminar, 3 May 2022.                                                                      |             |
|                                               | • <i>Moduli space of curves and abelian varieties</i> . Number Theory Learning Seminar, UC Berkeley, 23 Feb. 2022.                                                                      |             |
|                                               | • <i>Representations of totally disconnected groups</i> . Automorphic Representations Learning Seminar, UC Berkeley, 16 Feb. 2022.                                                      |             |
|                                               | • <i>Automorphic representations in the <math>L^2</math>-sense</i> . Automorphic Representations Learning Seminar, UC Berkeley, 2 Feb. 2022.                                            |             |
|                                               | • <i>Construction of Quot schemes</i> . Student Arithmetic Geometry Seminar, UC Berkeley, 24 Sept. 2021.                                                                                |             |
|                                               | • <i>Base change and the <math>q</math>-expansion principle</i> . Student Number Theory Seminar, University of California, Berkeley CA, 15 Sept. 2021.                                  |             |
|                                               | • <i>Modular forms and <math>q</math>-expansions in algebraic geometry</i> . Student Number Theory Seminar, University of California, Berkeley CA, 8 Sept. 2021.                        |             |
|                                               | • <i>Enumerating Hassett's decomposition of the moduli of curves</i> , with Daniel Gershenson and Elaine Hou. MAA Student Paper Session, MathFest, Chicago, IL, 2017.                   |             |
|                                               | • <i>Simplicial Complexes and Effective Divisors on <math>\overline{M}_{0,n}</math></i> , with Jocelyn Wang and Nick Wawrykow. MAA Student Paper Session, MathFest, Columbus, OH, 2016. |             |
| TECHNICAL<br>SKILLS                           | • John Alan Lewis Summer Research Fellowship, Yale                                                                                                                                      |             |
|                                               | • Fellow's Prize, Jonathan Edwards College, Yale                                                                                                                                        |             |
|                                               | • <i>Languages</i> : English, French, Python, C/C++, LaTeX, SageMath, MAGMA.                                                                                                            |             |